Keshav

Assignment 1:

1)

$9^3 = 9^3 = (3^2)^3 = 3^6$

$2^{32} = 2^{23}$

$16^4 = 16^4 = (2^4)^4 = 2^{16}$

$\boxed{3^6 = 3^6 \#}$

2)

$\Rightarrow$ 90% of 2 $= \sqrt{576} \div \sqrt{64} \times 729^{\frac{1}{3}}$

$= (24 \div 8) \times 9 = 3 \times 9 = 27$

Now, 90% of $x = 27$

$x = \frac{27 \times 100}{90} = 30$

$\boxed{\text{Ans} = 30 \#}$

3)

$(1/4) \times (4856 \times 0.5) \times 12$

$= \frac{1}{4} \times \frac{1}{2} \times 4856 \times 12$

$= \frac{4856 \times 12}{8}$

$\frac{12}{8} = \frac{3}{2}$ Now

$= \cancel{\frac{4856 \times 12}{8}}$ $4856 \times \frac{3}{2} = 2428 \times 3 = 7284$

$\boxed{\text{Answer} = 7284 \#}$

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4) $6153 \div \sqrt{x} \times 53 = 4028$

Now

$$\frac{6153 \times 53}{\sqrt{x}} = 4028$$

$$6153 \times 53 = 326109$$

$$\sqrt{x} = \frac{326109}{4028} = 81$$

$$x = 81^2 = 6561$$

$\boxed{\text{Answer} = 6561 \#}$

5) Price ↑ $16\frac{2}{3}\% = \frac{1}{6}$

Consumption ↓ $20\% = \frac{4}{5}$

New expenditure $= \frac{7}{6} \times \frac{4}{5} = \frac{28}{30} = \frac{14}{15}$

$= \frac{1}{15} = 6.67\%$

$\boxed{\text{Answer} = 6\frac{2}{3}\% \text{ decrease} \#}$

c) $62\frac{2}{3}\% = \frac{178}{3} \approx \frac{2}{3}$

Now $\frac{1}{3}x = 5670$

$x = 5670 \times 3 = 17010$

Answer $= 17010$ #

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6)

Volume $\propto r^2 h$

Radius ↑ 40% = 1.4

Height ↓ $37\frac{1}{2}\%$ = 62.5% = 0.625

New volume = $(1.4)^2 \times 0.625$

= $1.96 \times 0.625 = 1.225$

Increase

= 22.5%

$\boxed{\text{Answer} = 22.5\% \text{ Increase}}$ #

7)

→ $62\frac{2}{3}\% = \frac{188}{3} = \frac{2}{3}$

Remaining = $\frac{1}{3}$

$\frac{1}{3}x = 5620$

$x = 5620 \times 3 = 16860$

$\boxed{\text{Answer} = 16860}$ #

8)

Net Change = $\frac{\text{Change in tax}}{100 - \text{tax}}$

tax ↑ 17%

Net income ↓ 3%

$\frac{17}{100-x} = 3$

$17 = 3(100-x)$

$17 = 300 - 3x$

$x = 94.33\%$

$\boxed{\text{Answer} = 94.33\%}$ #

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9)

⇒ Rabbit = 4 legs, Pigeon = 2 legs.

let rabbits = r

Heads: $r + p = 280$

legs: $4r + 2p = 820$

Multiply first by 2:

$2r + 2p = 560$

Subtract:

$(4r + 2p) - (2r + 2p) = 820 - 560$

$2r = 260 \Rightarrow r = 130$

$\boxed{\text{Answer} = 130 \text{ rabbits}}$

(b)

Marks difference: $64 + 32 = 96$

Percentage diff. $48 - 36 = 12\%$

Total marks: $\frac{96 \times 100}{12} = 800$

Passing marks: $= 36\% + 32 = 288 + 32 = 320$

Passing % :- $\frac{320}{800} \times 100 = 40\%$

$\boxed{\text{Answer} = 40\%}$

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11)

$118.3 \div \frac{1}{0.7} \div 132 = 18\%$ of $x + 25\%$ of 7

$118.3 \times 0.7 = 82.81$

$\frac{82.81}{132} = 0.627 = 0.63$

Right side:

25% of $7 = 1.75$

$0.18x + 1.75 = 0.63$

$0.18x = -1.12 = x = -6$

$\boxed{\text{Answer} = -6 \ \#}$

12)

$\sqrt{625} \times \sqrt{6241} - 768 \times 3 + \sqrt{15129} + 606 \times 2$

Trick

$\sqrt{625} = 25$, $\sqrt{6241} = 79$ and $\sqrt{15129} = 123$

$25 \times 79 = 1975$

$768 \times 3 = 2304$

$606 \times 2 = 212$

$1975 - 2304 + 123 + 212 = 6$

$\boxed{\text{Answer} = 6 \ \#}$

13)

$x + 13 \times 50 = 420 + 45\%$ of $800 + 220$

$13 \times 50 = 650$

45% of $800 = 360$

$x + 650 = 420 + 360 + 220 = 1000$

$x = 350$

$\boxed{\text{Answer} = 350 \ \#}$

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14)

$\Rightarrow$ $\sqrt{225} \times 28 + 203 \times 7 = (?)^2$

$\sqrt{225} \times 28 = 98$

$203 \times 7 = 1421$

$980 + 1421 = 2401 = 49^2$

$\boxed{\text{Answer} = 49 \#}$

15)

$\Rightarrow$ 65% of 500 + 2x% of 350 = 500

65% of 500 = 325

$325 + 3.5x = 500$

$x = 50$

$\boxed{\text{Answer} = 50 \#}$

16)

$\Rightarrow$ $\dfrac{\{(18.99)^2 - 27.05^2 + (1394.95 + 59.85)\}}{x^2} = 39$

$a^2 - b^2 = (a-b)(a+b)$

$(18.99 - 27.05)(18.99 + 27.05) = (-8.06)(46.04)$

$= -371$

$1394.95 + 59.85 = 1459.8$

$1088.8 \div x = 39 \rightarrow x = 28$

$\boxed{\text{Answer} = 28}$

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17)

⇒ 21.97% of 899.5 + x + 39.77% of 309.76 = 481

21.97% = 22% ⇒ 899 × 0.22 = 198

39.77% = 40% ⇒ 309.76 × 0.40 = 124

198 + 124 = 322

x = 481 − 322 = 159 #

Answer = 159 #

18)

⇒ 40% sold ⇒ 60% left = 420

x × 0.6 = 420.

x = 700

Answer = 700 apples #

19)

⇒ 1136, 7636, 11628 (winner)

total = 20400

$\frac{11628}{20400}$ = 57% #

Answer = 57 #

20)

⇒ cost = 6150

6% rebate:

6150 × 0.94 = 6231

+10% tax

6231 × 110 = 6876

Answer = Rs. 6876.10 #

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22)

⇒ 8-year students = 48

above = $\frac{2}{3} \times 48 = 32$

Total event below:

$48 + 32 = 80$

$80\% = 80 \Rightarrow$ total $= 100$

$\boxed{\text{Answer} = 100 \#}$

23)

⇒ Correct $= \frac{5}{3}$, wrong $= \frac{3}{5}$

Error:

$$\frac{\left(\frac{5}{3} - \frac{3}{5}\right)}{\frac{5}{3}} \times 100$$

$$\frac{16/15}{5/3} = \frac{16}{25} = 64\%$$

$\boxed{\text{Answer } 64\% \#}$

24)

⇒ $x = 1/20$, $y = 1/12$

4 days x:

$4 \times \frac{1}{20} = \frac{1}{5}$

Remaining $= 4/5$

Together

$\frac{1}{20} + \frac{1}{12} = \frac{8}{60} = \frac{2}{15}$

$\frac{4/5}{2/15} = 6$ days.

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total 446 = 10

$\boxed{\text{Answer} = 60 \text{ days}}$

25)

→ A = 20, B = 30, C = 60

3-day cycle:

Day 1: A = 1/20

Day 2: A = 1/20

Days 3: A+B+C = 1/20 + 1/30 + 1/60 = 1/10

= 1/20 + 1/20 + 1/10 = 1/5

15 days = 1 works

$\boxed{\text{Answer} = 15 \text{ days}}$

26)

⇒ (A+B) = 1/12, (B+C) = 1/15, (C+A) = 1/20

2(A+B+C) = 1/12 + 1/15 + 1/20

= 12/60 = 1/5

A+B+C = 1/10

$\boxed{\text{Time} = 10 \text{ days}}$

27)

→ P = 18, a_0 = 1/12

= 1/8 + 1/12 = 5/24

4-cycles

= 20/24 = 5/6

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Remaining = 1/6

Next → P work:

$\frac{1}{6} \div \frac{1}{8} = \frac{4}{3}$ hr = 1 hr 20 min

Total = 7h 20 min

Start 5:00 am → 6:20 pm

$\boxed{\text{Answer} = 6:20 \text{ pm}}$ #

28)

→ $A = \frac{1}{15}$, $B = \frac{1}{25}$

Let total days = n

B worked (n−7) days

$\frac{n}{15} + \frac{n-7}{25} = 1$

LCM 75: $8n + 3(n-7) = 75$

$8n - 21 = 75 \rightarrow 2n = 12$

$\boxed{\text{Answer} = 12 \text{ days}}$ #

29) Man 1 = w/5

Man 2 = w/10

$\frac{w}{5} + \frac{w}{10} = \frac{1500}{90}$

Given: $\frac{w}{5} = \frac{1500}{90} - 10$

$\frac{1500}{90} = \frac{1500}{90} - 10$

$w = 900$

$\boxed{\text{Answer} = 900 \text{ bricks}}$ #

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30)

$\Rightarrow$ $6M + 8B = 1/10$

$26M + 24B = 1/2$

multiply first $\times 3$:

$18M + 24B = 3/10$

Subtract: $8M + 24B = 1/2 - 3/10 = 1/5$

$M = 1/40$, $B = 1/80$

$15M + 20B$:

$= \frac{15}{40} + \frac{20}{80} = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$

Time $= \frac{8}{5} = 1.6$ days

$\boxed{\text{Answer} = 1.6 \text{ days} \#}$

31)

$\Rightarrow$ $A = \frac{1}{4}$

$A + C = \frac{1}{3}$

$A + C = \frac{1}{2}$

C: $(A+C) - A = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$

B: $(B+C) - C = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$

Ans:

$\boxed{B \text{ alone} = 12 \text{ days} \#}$

32)

B (10 days)

$10 \times \frac{1}{15} = \frac{2}{3}$

remaining $\frac{1}{3}$

A time: $1/3 / 1/18 = 6$

$\boxed{\text{Answer} = 6 \text{ days} \#}$

3b)

$\Rightarrow$ $4m + 6w = 1/8$

$3m + 7w = \frac{1}{10}$

Solve:

$m = \frac{1}{40}$, $w = \frac{1}{80}$

10w:

$= \frac{10}{80} = 1/8$

Time = 8 days

$\boxed{\text{Answer} = 8 \text{ days}}$

3c)

$x = 1/20$, $y = \frac{1}{12}$

4 days x:

$4 \times \frac{1}{20} = \frac{1}{5}$

Remaining = 4/5

$\frac{1}{20} + \frac{1}{12} = \frac{8}{60} = \frac{2}{15}$

$\frac{4/5}{2/15} = 6$ days

total = 4+6=10

$\boxed{\text{Answer} = 10 \text{ days}}$

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35)

$\Rightarrow$ $A = B + C$

$A + B = \frac{1}{15}$, $C = \frac{1}{50}$

A:

$2A - C = \frac{1}{10}$

$2A = \frac{1}{10} + \frac{1}{50} = \frac{3}{25} \Rightarrow A = \frac{3}{50}$

B:

$B = A - C = \frac{3}{50} - \frac{1}{50} = \frac{1}{25}$

$\boxed{\text{Answer} = 25 \text{ days} \#}$

36) A does 80% in 20 days $\Rightarrow$ Rate = 25 days

Remaining = 20%

(A+B) rate:

$\frac{\frac{1}{5}}{3} = \frac{1}{15}$

B: $\frac{1}{15} - \frac{1}{25} = \frac{2}{75}$

Time: $= \frac{75}{2} = 37.5$

$\boxed{\text{Answer} = 37.5 \text{ days} \#}$

37)

$\Rightarrow$ B (10 days): $10 \times \frac{1}{15} = \frac{2}{3}$

Remaining: $1/3$

A time: $\frac{1/3}{1/18} = 6$

$\boxed{\text{Answer} = 6 \text{ days} \#}$

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38)

⇒ Let A-h

$$\frac{1}{h} + \frac{1}{B} = \frac{1}{30}$$

Work done:

$$20 \times \frac{1}{30} = \frac{2}{3}$$

Remaining $= \frac{1}{3}$

$$20 \times \frac{1}{h} = \frac{1}{3}$$

$$\frac{1}{h} = \frac{1}{60}$$

$\boxed{\text{Answer} = 60 \text{ days}}$

39)

⇒ Ravi: $\frac{32}{6} = \frac{16}{3}$

Kumar: $\frac{40}{5} = 8$

Together: $= \frac{16}{3} + 8 = \frac{40}{3}$

Time: $110 \div \frac{40}{3} = 8.25$ hrs.

$\boxed{\text{Answer} = 8 \text{ hr } 15 \text{ min}}$

40)

⇒ $A + B + C = \frac{1}{14}$

10 days: $= \frac{10}{14} = \frac{5}{7}$

A: $\frac{1}{20} + \frac{1}{30} - \frac{1}{24} = \frac{1}{80}$

$\frac{3+8}{1/8} = 30$

$\boxed{\text{Answer} = 30}$

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41)

$\Rightarrow$ $20W = 1/10 \Rightarrow W = 1/320$

$16M = 1/15 \Rightarrow M = 1/240$

Ratio : $M : W = \frac{1/240}{1/320} = \frac{4}{3}$

$\boxed{\text{Answer} = 4:3 \#}$

42)

$\Rightarrow$ B+C: $= \frac{1}{4} + \frac{1}{12} = \frac{7}{30}$

3 days: $= \frac{7}{12}$

Remaining $= 5/12$

A: $= \frac{1}{24}$

Time $= \frac{5/12}{1/24} = 10$

$\boxed{\text{Answer} = 10 \text{ days} \#}$

43)

$\Rightarrow$ $A+B = 1/30$

$\frac{10}{30} + \frac{44}{B} = 1$

$\frac{44}{B} = \frac{14}{30}$

$B = \frac{1320}{14} = 60$

$\boxed{\text{Answer} = 60 \text{ days} \#}$

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44)

⇒ $B = 1/12$, $A = 1/6$

Together:

$$= \frac{1}{6} + \frac{1}{12} = \frac{1}{4}$$

$\boxed{\text{Answer} = 4 \text{ days } \#}$

45)

⇒ $\frac{1}{20} + \frac{1}{30} = \frac{5}{60} = \frac{1}{12}$

$\boxed{\text{Answer} = 12 \text{ min } \#}$

46)

⇒ 4 min:

$$4 \times (1/15 + 1/20) = 7/15$$

Remaining $= 8/15$

B:

$$= 1/20$$

Time:

$$= \frac{8/15}{1/20} = 10.67$$

Closest

10 min 20 sec

$\boxed{\text{Answer} = 10 \text{ min } 20 \text{ sec } \#}$

(47)

⇒ total = $\frac{1}{6}$

2 hr work = $1/5$

Remaining = $\frac{2}{5}$

A+B :

$= \frac{2/5}{7} = \frac{2}{21}$

C : $1/6 - 2/21 = 1/14$

$\boxed{\text{Answer} = 14 \text{ hours}}$

(48)

⇒ A always = $1/12$

B & C alternate

24 hrs

$\left(\frac{1}{12} + \frac{1}{15}\right) + \left(1/12 + 1/20\right) = 3/10$

$= \frac{20}{3} = 6.67$

$\boxed{\text{Answer} = 6.6 \text{ hours}}$